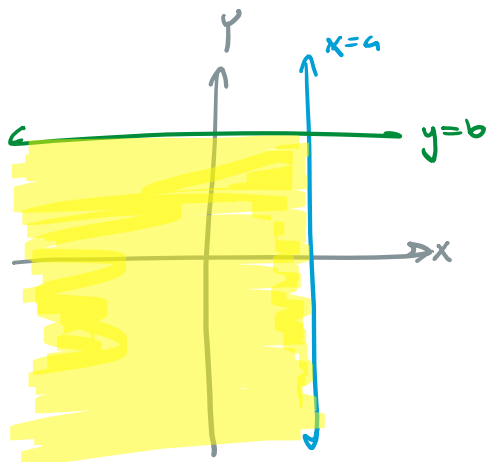


6.1: Joint Distribution Functions

Given two random variables X and Y , we say the **cumulative probability distribution function** of X and Y is $F(a, b) := P(X \leq a, Y \leq b)$ for any $a, b \in \mathbb{R}$



CONTINUOUS:

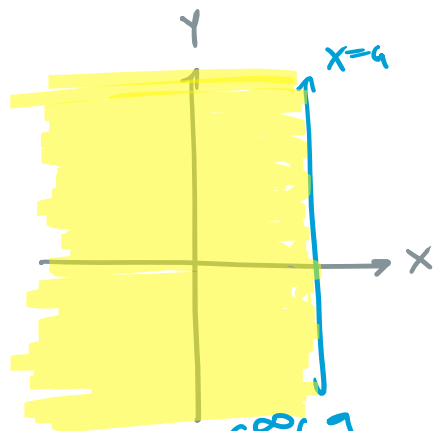
$$P(X \leq a, Y \leq b) = \int_{-\infty}^a \int_{-\infty}^b \underbrace{f_{XY}(x, y)}_{\text{JOINT DENSITY FUNCTION}} dy dx$$

DISCRETE:

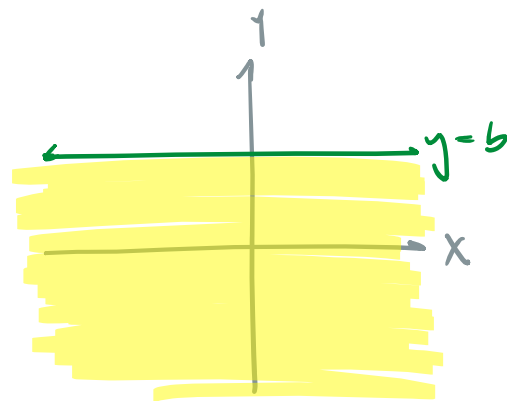
$$P(X \leq a, Y \leq b) = \sum_{x \leq a} \sum_{y \leq b} \underbrace{p_{XY}(x, y)}_{\text{JOINT POINT MASS FUNCTION}}$$

This quantity is often related to the **marginal distribution functions** for X and Y :

$$F_X(a) = P(x \leq a) = F(a, \infty) \text{ and } F_Y(b) = P(y \leq b) = F(\infty, b)$$

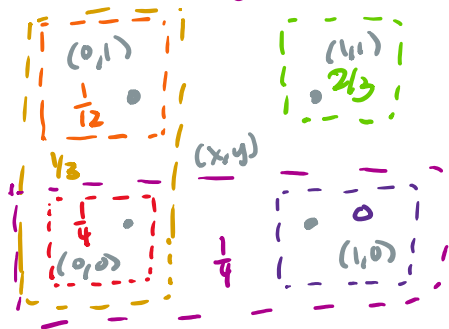


$$\begin{aligned} P(X \leq a) &= \int_{-\infty}^{\infty} \int_{-\infty}^a f_{XY}(x, y) dx dy \\ &= \sum_y \sum_{x \leq a} p_{XY}(x, y) \end{aligned}$$



$$\begin{aligned} P(Y \leq b) &= \int_{-\infty}^{\infty} \int_{-\infty}^b f_{XY}(x, y) dy dx \\ &= \sum_x \sum_{y \leq b} p_{XY}(x, y) \end{aligned}$$

Example: Consider X and Y with range $\{0, 1\}$ satisfying
 $F(0, 0) = \frac{1}{4}$, $F(0, 1) = \frac{1}{3}$, $F(1, 0) = \frac{1}{4}$, $F(1, 1) = 1$. Find F_X and F_Y



$$1 = p_{xy}(0,0) + p_{xy}(0,1) + p_{xy}(1,0) + p_{xy}(1,1)$$

$$1 = \frac{1}{4} + \frac{1}{3} + 0 + \frac{2}{3}$$

$$\frac{1}{4} = F(0,0) = P(X \leq 0, Y \leq 0) = P(X=0, Y=0) = p_{xy}(0,0)$$

$$\frac{1}{3} = F(0,1) = P(X \leq 0, Y \leq 1) = P(X=0, Y=\text{any}) = P(X=0) = p_{xy}(0,1) + p_{xy}(0,0)$$

$$\frac{1}{4} = F(1,0) = P(X \leq 1, Y \leq 0) = P(X=\text{any}, Y=0) = P(Y=0) = p_{xy}(0,0) + p_{xy}(1,0)$$

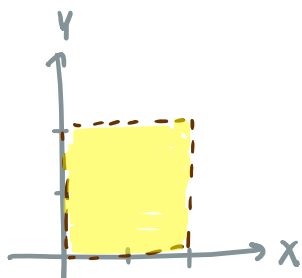
$$F_X(a) = P(X \leq a) = P(X \leq a, Y=0) + P(X \leq a, Y=1)$$

$$= \begin{cases} 0 & a < 0 \\ \frac{1}{3} & 0 \leq a < 1 \\ 1 & a \geq 1 \end{cases}$$

$$F_Y(b) = P(Y \leq b) = P(Y \leq b, X=0) + P(Y \leq b, X=1)$$

$$= \begin{cases} 0 & b < 0 \\ \frac{1}{4} & 0 \leq b < 1 \\ 1 & b \geq 1 \end{cases}$$

Example: Let $f(x, y) = kxy^2$ for $0 \leq x \leq 2, 0 \leq y \leq 2$ and zero elsewhere. Find $P(X + Y \leq 2)$

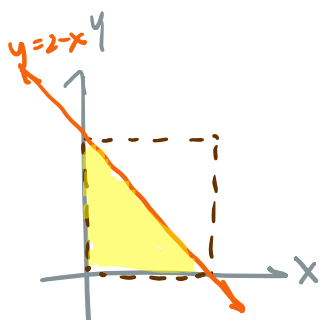


Find k :

$$1 = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{XY}(x, y) dA = \int_0^2 \int_0^2 k \cdot x y^2 \cdot dy dx$$

$$= k \cdot \int_0^2 x dx \cdot \int_0^2 y^2 dy = k \cdot \left. \frac{1}{2} x^2 \right|_0^2 \cdot \left. \frac{1}{3} y^3 \right|_0^2 = k \cdot 2 \cdot \frac{8}{3} = 1$$

$$k = \frac{3}{16}$$



$$P(X + Y \leq 2) = P(Y \leq 2 - X) = \int_0^2 \int_0^{2-x} f_{XY}(x, y) dy dx$$

$$= \int_0^2 \int_0^{2-x} \frac{3}{16} x y^2 dy dx = \frac{3}{16} \int_0^2 x \cdot \left. \frac{1}{3} y^3 \right|_0^{2-x} dx$$

$$= \frac{1}{16} \int_0^2 x (2-x)^3 dx \quad \left(\begin{array}{ll} u = 2-x & x=u \rightarrow u=2 \\ du = -dx & x=2 \leftarrow u=0 \end{array} \right)$$

$$= \frac{1}{16} \int_2^0 (2-u) \cdot u^3 \cdot (-du)$$

$$= \frac{1}{16} \int_0^2 2u^3 - u^4 du$$

$$= \frac{1}{16} \left[\frac{1}{2} u^4 - \frac{1}{5} u^5 \right] \Big|_0^2$$

$$= \frac{1}{16} \left[8 - \frac{32}{5} \right]$$

$$= \frac{1}{16} \cdot \frac{8}{5} = \boxed{\frac{1}{10}}$$

Exactly how we did with one random variable, we can define expectations of functions of joint distributions as well:

CONTINUOUS:

$$E[g(X,Y)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x,y) f_{XY}(x,y) dA$$

DISCRETE:

$$E[g(X,Y)] = \sum_x \sum_y g(x,y) p_{XY}(x,y)$$

Example: Let $f(x,y) = e^{-\sqrt{x^2+y^2}}$ and consider X,Y with $F(a,b) = \int_{-\infty}^a \int_{-\infty}^b k \cdot f(s,t) dt ds$

Solve for $E[XY]$

Find k :

$$\begin{aligned} 1 &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} k \cdot e^{-\sqrt{x^2+y^2}} dA = \int_0^{2\pi} \int_0^{\infty} k \cdot e^{-r} \cdot r dr d\theta \\ &= k \cdot \int_0^{2\pi} d\theta \cdot \int_0^{\infty} r e^{-r} dr = k \cdot \theta \Big|_0^{2\pi} \cdot \left[-r e^{-r} - e^{-r} \right]_0^{\infty} \\ &= k \cdot 2\pi \cdot \left[(0-0) - (0-1) \right] = k \cdot 2\pi \cdot 1 = 1 \longrightarrow k = \frac{1}{2\pi} \end{aligned}$$

r	e^{-r}
1	$-e^{-r}$
0	e^{-r}

$$E[XY] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xy \cdot f_{XY}(x,y) dA = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xy \cdot \frac{1}{2\pi} e^{-\sqrt{x^2+y^2}} dA$$

$$\begin{aligned} &= \frac{1}{2\pi} \int_0^{2\pi} \int_0^{\infty} r \cos \theta \cdot r \sin \theta \cdot e^{-r} \cdot r dr d\theta = \frac{1}{2\pi} \int_0^{2\pi} \sin \theta \cos \theta d\theta \cdot \int_0^{\infty} r^3 e^{-r} dr \\ &= \frac{1}{2\pi} \cdot \left[\frac{1}{2} \sin^2 \theta \right]_0^{2\pi} \cdot \left[\int_0^{\infty} r^3 e^{-r} dr \right] \\ &= \frac{1}{2\pi} \left[\frac{1}{2} \cdot 0 - \frac{1}{2} \cdot 0 \right] \cdot (\text{WRTO LATER}) = \boxed{0} \end{aligned}$$

Example: A game is played with n^2 pegs arranged in n rows and n columns. (Each peg is labeled (i, j) with $1 \leq i, j \leq n$ for convenience). The contestant throws a ring until it lands on exactly one peg, and they are rewarded with $\$ij$. If we know the probability of a the ring landing on peg (i, j) is ki^2j^2 for each peg, what is the expected payout of the game?

$(1,5) \dots (5,5)$

$(1,1) \dots (5,1)$

$(4,3)$ Pay = $\$4 \cdot 3 = \12

$$\text{Find } k: 1 = \sum_{i=1}^n \sum_{j=1}^n P_{IJ}(i, j) = \sum_{i=1}^n \sum_{j=1}^n k \cdot i^2 \cdot j^2$$

$$= k \cdot \left(\sum_{i=1}^n i^2 \right) \left(\sum_{j=1}^n j^2 \right) = k \cdot \frac{n(n+1)(2n+1)}{6} \cdot \frac{n(n+1)(2n+1)}{6}$$

$$\left(\sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6} \right)$$

$$\left(\sum_{i=1}^n i^3 = \frac{n^2(n+1)^2}{4} \right)$$

$$k = \frac{36}{n^2(n+1)^2(2n+1)^2}$$

$$E[IJ] = \sum_{i=1}^n \sum_{j=1}^n ij \cdot P_{IJ}(i, j) = \sum_{i=1}^n \sum_{j=1}^n ij \cdot k \cdot i^2 \cdot j^2$$

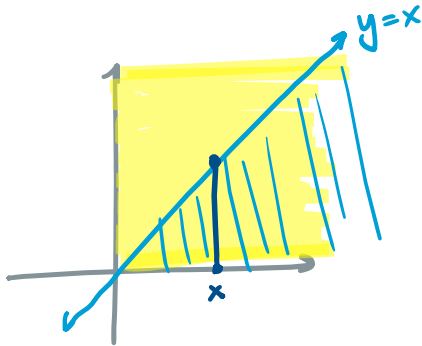
$$= k \cdot \left(\sum_{i=1}^n i^3 \right) \left(\sum_{j=1}^n j^3 \right) = k \cdot \frac{n^2(n+1)^2}{4} \cdot \frac{n^2(n+1)^2}{4}$$

$$= \frac{36}{n^2(n+1)^2(2n+1)^2} \cdot \frac{n^4(n+1)^4}{16}$$

$$\boxed{= \frac{9n^2(n+1)^2}{4(2n+1)^2}} \xrightarrow{"n \rightarrow \infty"} \frac{9}{16} n^2$$

Example: Let X, Y have joint density $f_{XY}(x, y) = \begin{cases} abe^{-ax-by} & x, y \geq 0 \\ 0 & \text{else} \end{cases}$ for $a, b > 0$.

What is $P(X > Y)$?



$$I = \int_0^{\infty} \int_0^{\infty} abe^{-ax-by} dx dy \quad (\text{TPNST ME})$$

$$\begin{aligned} P(X > Y) &= \int_0^{\infty} \int_0^x abe^{-ax-by} dy dx \\ &= \int_0^{\infty} ae^{-ax} \cdot (-e^{-by}) \Big|_0^x dx \\ &= \int_0^{\infty} ae^{-ax} \cdot [-e^{-bx} + 1] dx \\ &= \int_0^{\infty} ae^{-ax} - ae^{-(a+b)x} dx \\ &= -e^{-ax} + \frac{a}{a+b} e^{-(a+b)x} \Big|_0^{\infty} \\ &= (0 + 0) - (-1 + \frac{a}{a+b}) \\ &= \boxed{\frac{b}{a+b}} \end{aligned}$$